ASTEROSEISMOLOGY

Stars also experience seismic waves due to their massive energy. Although their sound cannot travel, with the right technology, you can hear the the symphony of the stars.

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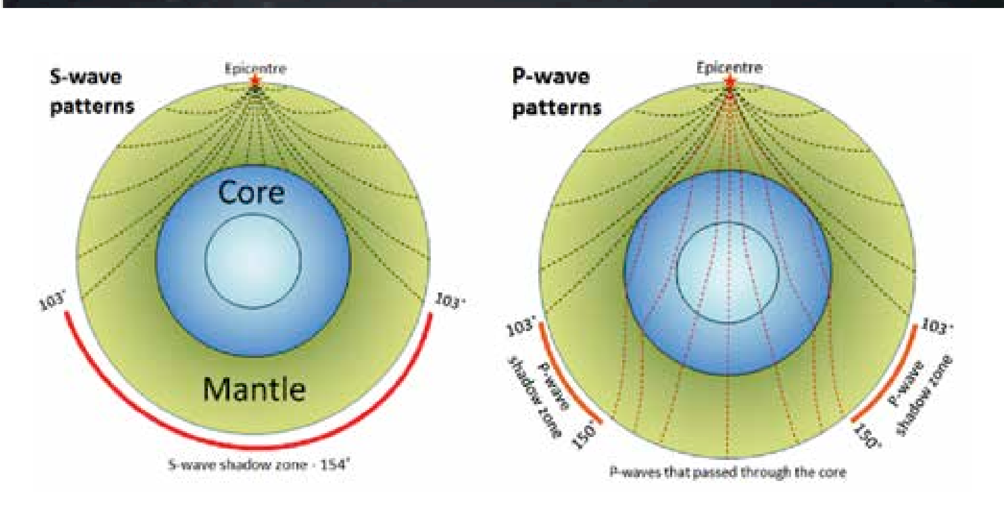
The word Asteroseismology can easily be broken down into its constituents to reveal its meaning. 'Astero' most probably derives from 'astro' which relates to stars and seismology is the study of the propagation of elastic waves, 'quakes', through the internal structure of a large geological body like the Earth. Asteroseismology hence is the study of seismic waves in stars. By studying these wave induced oscillations, scientists can decode the internal structure of stars, making asteroseismology an invaluable tool for astronomy. Seismic waves have long been used to map the structure of our planet's interior. Waves that travel in the interior of the Earth are of two types - p (pressure) waves and s (shear) waves. Since both waves are different in nature, they travel differently in liquids and solids. In the event of an Earthquake, both P and S waves are generated in the Earth's body and by measuring the intensity of these waves geologists have resolved the internal structure of the Earth upto a resolution of a few hundred kilometers. Asteroseismology uses a similar principle to study what goes on inside stars. In this article we discuss how this nascent field helps physicists and astronomers obtain invaluable data about stellar bodies in our cosmos. We highlight how asteroseismology allows us to 'hear' stars and what that truly means. The stars across galaxies pulsate and throb like a chaotic orchestra. What if we can tune into the cosmic concert and listen to the symphony of stars? You better have a good seat for this one.



How S and P waves resolve the Earth's internal structure owing to their respective abilities or inability to pass through solid and liquid layers